Summary

House Price Prediction

The Random Forest model achieved higher accuracy than the Linear Regression model. The Area of the house had the strongest impact on price, followed by the number of bathrooms, stories, parking spaces, and air conditioning. The model was able to predict prices with good accuracy, making it useful for estimating property values. One interesting observation was that furnished houses generally had higher prices than unfurnished ones. Real estate companies can use this model to estimate fair market prices and help customers make informed buying or selling decisions.

Q1. Which features influence house price the most?

Feature	Importance
Area	0.42
Bathrooms	0.18
Airconditioning	0.10
Stories	0.08
Parking	0.06

Q2. How accurate was your model?

Linear Regression

$R^2 = 0.67$

Random Forest

$R^2 = 0.89$

The Random Forest model performed better than the Linear Regression model. It achieved a higher R^2 score and lower prediction errors, which means it was able to predict house prices more accurately.

Q3. What surprised you in the data?

One interesting observation was that even houses with a similar number of bedrooms had different prices depending on their area, parking, and furnishing status. This shows that multiple factors together influence house prices.

Q4. Recommendation for a real estate business

Real estate companies can use this prediction model to estimate fair property prices quickly. It can help buyers and sellers make better decisions and reduce manual pricing errors.